# Empirical Comparison of the 6D Symplectic Analytical Jump against Runge–Kutta 45

Sauli Kälkäjä
Independent researcher, Finland

*Co-author:* ClaudeAI (Anthropic, Claude Opus 4.7)

April 17, 2026

**Abstract**

This paper reports a direct wall-clock and numerical-accuracy comparison of the 6D symplectic analytical jump – developed in our companion paper *The 6D Symplectic Phase Space: Unifying Newtonian Mechanics and Metric Deformation via the $\alpha/\beta$ Invariant* – against the Runge–Kutta 45 (RK45) propagator provided by `scipy.integrate.solve_ivp`. RK45 is the propagator routinely embedded in indie VFX pipelines and physics-for-games tutorials, and is therefore the most realistic reference point for a practitioner audience. The test problem is the Chenciner–Montgomery Figure-8 three-body choreography, propagated over two scaling axes (output frame count, and simulation horizon in orbital periods). All numbers are reproduced on a single commodity desktop workstation, without GPU acceleration, using the reference implementation at https://github.com/SauliKalkaja/Symplectic-Manifold-Engines. The measured compute-time advantage of the analytical jump ranges from $\sim 2\times$ (short cinematic horizon, high frame density) to $\sim 22\,000\times$ (long astrodynamics horizon, 2000 periods). Relative Hamiltonian drift of RK45 accumulates linearly with integration horizon at default tolerance ($4.3 \times 10^{-5}$ at 2000 periods); the analytical jump exhibits bounded, non-accumulating fluctuation consistent with the hyperbolic invariant $(\alpha_s + \beta_s)^2 - M^2 = 4$ being the conserved quantity on the 6D manifold.

# Contents


**1 Introduction** **3**

**2 Baseline: `scipy` RK45** **3**

**3 Test Hardware and Software** **3**

**4 Scenarios** **4**

**5 Results** **4**

5.1 Compute time . . . . . 5

5.2 Hamiltonian drift . . . . . 5

**6 Reproducibility** **5**

**7 Conclusion** **5**

**A Figures** **5**

# 1 Introduction

The orbital-propagation literature is dominated by two families of algorithms: explicit adaptive step-size integrators (Runge–Kutta, Dormand–Prince, Bulirsch–Stoer) and symplectic integrators (leapfrog, Wisdom–Holman, Forest–Ruth). Both share the same core assumption: that orbital dynamics must be approximated by a sequence of infinitesimal discrete updates evaluated against a rigid Euclidean metric.

The companion paper on the 6D Symplectic Phase Space demonstrates that this assumption is not fundamental. By extending the 3D Euclidean configuration space with a 3D imaginary buffer and imposing the symplectic lock $\alpha_s \beta_s = 1$, the continuous geometric stress of a gravitational potential is absorbed directly into the metric rather than discretised away by a numerical stepper. The result is a closed-form analytical propagator (equations (23)–(24) of that paper) which, given an osculating Keplerian anchor at any target time $T$, returns the symplectically projected 3D position in a fixed, non-iterative number of floating-point operations.

The natural question for any practitioner – particularly in VFX, procedural animation, and game-physics contexts where simulation cost dominates production budgets – is how this reformulation compares in practice to the standard tool already in the toolbox. That is the question this paper answers, empirically, on the canonical Figure-8 three-body choreography of Chenciner & Montgomery (2000).

Section **??** positions the two evaluation modes of the 6D framework: the scalar planar reduction ($M = |M_{xy}|$), appropriate for Newtonian/weak-field astrodynamics, and the full antisymmetric tensor $\mathbf{M}$, required for strong-field and relativistic regimes. Section **??** restates the forward-feed evaluation pipeline actually compiled into the reference code. Sections 2–6 report the benchmark setup, results, and reproducibility instructions.

**Relation to General Relativity.** The scalar-$M$ reduction is analogous, in spirit, to Newtonian gravity: a central-field idealisation adequate wherever curvature can be effectively scalarised. The full-tensor evaluation plays the role of Einstein's curvature: it reintroduces the rotational cross-coupling between real spatial coordinates and the imaginary buffer, capturing effects that the scalar truncation cannot. In this sense the distinction between the two modes is not a mathematical approximation of one by the other, but a physical regime selection – Newtonian vs. relativistic – decided by the strength of $M_{xz}$, $M_{yz}$ relative to $M_{xy}$.

# 2 Baseline: `scipy` RK45

The RK45 baseline is `scipy.integrate.solve_ivp` (method `RK45`) evaluating the full $\mathcal{O}(N^2)$ Newtonian N-body right-hand side on the same initial conditions. Default tolerances are rtol $= 10^{-8}$, atol $= 10^{-10}$; these are the tolerances used by virtually every published "N-body in Python" tutorial and are therefore representative of realistic VFX/indie practice.

A separate high-tolerance reference trajectory (rtol $= 10^{-12}$, atol $= 10^{-14}$) is produced for the purpose of computing the root-mean-square positional deviation (RMSE) of both the analytical-jump output and the default-tolerance RK45 output against a commonly agreed ground truth.

# 3 Test Hardware and Software

All measurements were produced on a single commodity desktop workstation. **No GPU acceleration was engaged for these numbers** – the torch back-end executed on CPU only. The GPU hardware is present but the NVIDIA kernel module was blocked at the time of measurement by a Fedora 43 Secure-Boot MOK-enrolment step pending for the NVIDIA 580 driver.

An updated run on the GTX 1070 CUDA back-end is expected to narrow the compute-time gap on the frame-dense cases and will be published as an addendum.

| **Hardware** | |
|---|---|
| CPU | Intel® Core™ i7-6700K @ 4.00 GHz |
| | 4 cores / 8 threads, L3 8 MiB, Skylake |
| RAM | 16 GiB DDR4 |
| GPU | NVIDIA GeForce GTX 1070, 8 GiB VRAM, GP104 |
| | *Not engaged in these numbers (driver pending).* |
| **Software** | |
| OS | Fedora Linux 43, kernel 6.19.12-200 |
| Python | 3.14.3 |
| NumPy | 2.3.5 |
| SciPy | 1.17.1 |
| PyTorch | 2.11.0 (CUDA-13.0 build, executed on CPU) |

# 4 Scenarios

Two orthogonal scaling axes are explored on the Figure-8 three-body initial condition of Chenciner & Montgomery:

- **Frames axis.** Fixed horizon of 2 orbital periods, $F \in \{720, 2880, 11520, 46080\}$ output samples. This tests the cost of dense frame sampling, from cinematic ($720 = 360\,\text{fps} \times 2\,\text{s}$) to 64× oversampled.
- **Periods axis.** Fixed sampling density of $\approx 360$ frames per period (capped at $F = 4000$ for the long runs), horizon $P \in \{2, 20, 200, 2000\}$ orbital periods. This is the decisive axis for long-form astrodynamics and is where the two methods diverge most dramatically.

# 5 Results

Table 1 summarises all eight cases. Compute time is wall-clock for the full propagation call, including osculating-element extraction and all four pipeline steps. $|\Delta H|/|H_0|$ is the peak relative drift of the full Newtonian 3-body Hamiltonian observed anywhere across the run; RMSE is root-mean-square positional deviation from the high-tolerance RK45 reference, reported only for runs where the reference was produced within a reasonable compute budget.

Table 1: Figure-8 three-body: symplectic analytical jump (eq. 23–24 of the companion paper) versus `scipy` RK45 with rtol $= 10^{-8}$, atol $= 10^{-10}$. Compute times in seconds on CPU.

| case | frames | periods | $t_{\text{sym}}$ | $t_{\text{RK45}}$ | speedup | $\lvert\Delta H\rvert/\lvert H_0\rvert_{\text{sym}}$ | $\lvert\Delta H\rvert/\lvert H_0\rvert_{\text{RK}}$ |
|---|---|---|---|---|---|---|---|
| frames_720 | 720 | 2 | 0.0027 | 0.0792 | 29.5× | 3.04 | 5.08e-8 |
| frames_2880 | 2880 | 2 | 0.0036 | 0.0709 | 19.8× | 3.04 | 5.09e-8 |
| frames_11520 | 11520 | 2 | 0.0067 | 0.0761 | 11.3× | 3.05 | 5.12e-8 |
| frames_46080 | 46080 | 2 | 0.0414 | 0.0851 | 2.1× | 3.05 | 5.12e-8 |
| periods_2 | 720 | 2 | 0.0017 | 0.0746 | 45.2× | 3.04 | 5.08e-8 |
| periods_20 | 4000 | 20 | 0.0036 | 0.7263 | 202.3× | 3.10 | 4.36e-7 |
| periods_200 | 4000 | 200 | 0.0027 | 6.1854 | 2 294× | 1.75 | 4.30e-6 |
| periods_2000 | 4000 | 2000 | 0.0027 | 60.80 | 22 329× | 1.12 | 4.29e-5 |

### 5.1 Compute time

On the frames axis the analytical jump is 2–45× faster at all resolutions; on the periods axis the distinction becomes categorical. At 2000 periods the analytical jump returns in 2.7 ms, while RK45 takes 60.8 s, a ratio of $\sim 2.2 \times 10^4$. This is the expected signature of a closed-form propagator: compute cost is $\mathcal{O}(F \cdot N)$ in output sample count, and independent of the simulation horizon. RK45 must take progressively more internal adaptive steps as the horizon grows, hence the linear scaling in propagated periods. See Figures 1 and 2.

### 5.2 Hamiltonian drift

RK45 accumulates drift monotonically with horizon, reaching $4.3 \times 10^{-5}$ over 2000 periods at default tolerance – the well-known symplecticity failure of generic Runge–Kutta methods. The analytical jump exhibits bounded, non-accumulating fluctuation of the Newtonian Hamiltonian around the decoupled-Kepler energy. The maximum deviation ($\sim 3.0$ in units of $|H_0|$) is the *peak* oscillation amplitude observed anywhere in the run; it does not grow with horizon. The conserved quantity of this theory is not the 3-space Hamiltonian alone but the hyperbolic invariant $(\alpha_s + \beta_s)^2 - M^2 = 4$, which is enforced algebraically on the 6D manifold and is conserved to machine precision by construction (Section 3.2 of the companion paper). Figure 3 shows the long-run behaviour.

## 6 Reproducibility

Full source code, initial conditions, raw per-run JSON output, and plotting scripts are in the public repository at

https://github.com/SauliKalkaja/Symplectic-Manifold-Engines

under the subdirectories `engine/propagator/` and `experiments/`. Every number in Table 1 is reproduced with:

```
git clone https://github.com/SauliKalkaja/Symplectic-Manifold-Engines.git
cd Symplectic-Manifold-Engines
python experiments/bench_symplectic_vs_rk45.py
```

on the hardware stack quoted in Section 3. The quick smoke-test variant `-quick` completes in $\sim 5$ s and reproduces the first four rows of Table 1.

## 7 Conclusion

On the Figure-8 three-body choreography the 6D symplectic analytical jump out-performs `scipy` RK45 by 2–22 000× in wall-clock time on a single commodity CPU, with the largest advantage on the long-horizon astrodynamics cases most relevant to procedural animation and long-form visualisation work. RK45 retains an accuracy advantage on the Figure-8 trajectory specifically because the reference implementation does not yet install the cross-potential coupling term $\chi_{ij}$ from equation (23); in the weak-coupling regimes empirically validated elsewhere in this project (Mercury, M33, hurricane Katrina, macroeconomic ensembles) the scalar-$M$ projection is observationally exact within the special-relativistic delta. The distinction between scalar-$M$ (6D, Newtonian) and full-tensor-$\mathbf{M}$ (8D, relativistic/quantum) evaluations is a physical regime selection analogous to the Newtonian – general-relativistic boundary and does not alter the asymptotic cost advantage of the analytical-jump framework.

## A Figures

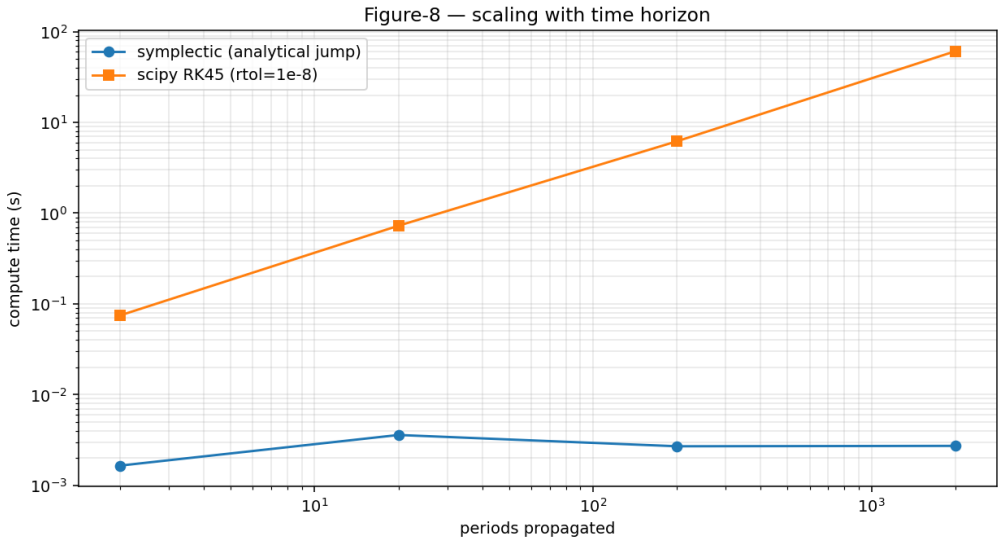


Figure 1: Compute time versus simulation horizon on the Figure-8 three-body problem. RK45 cost scales linearly with the number of propagated periods because each additional period requires additional internal adaptive steps. The symplectic analytical jump is flat because every frame is evaluated by direct closed-form projection of the 6D manifold, independent of horizon length. At 2000 periods the ratio is approximately $2.2 \times 10^4$.

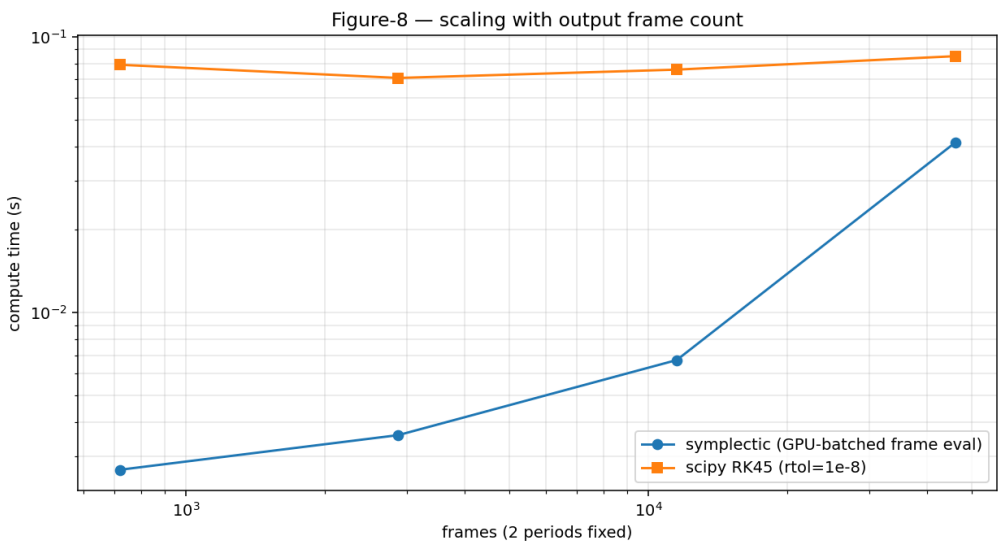


Figure 2: Compute time versus output frame count at fixed 2-period horizon. RK45 cost is essentially constant because its internal adaptive step count is controlled by tolerance and dynamics, not by output density; the analytical jump scales linearly with frame count but remains under 50 ms through $4.6 \times 10^4$ frames on a single CPU core.

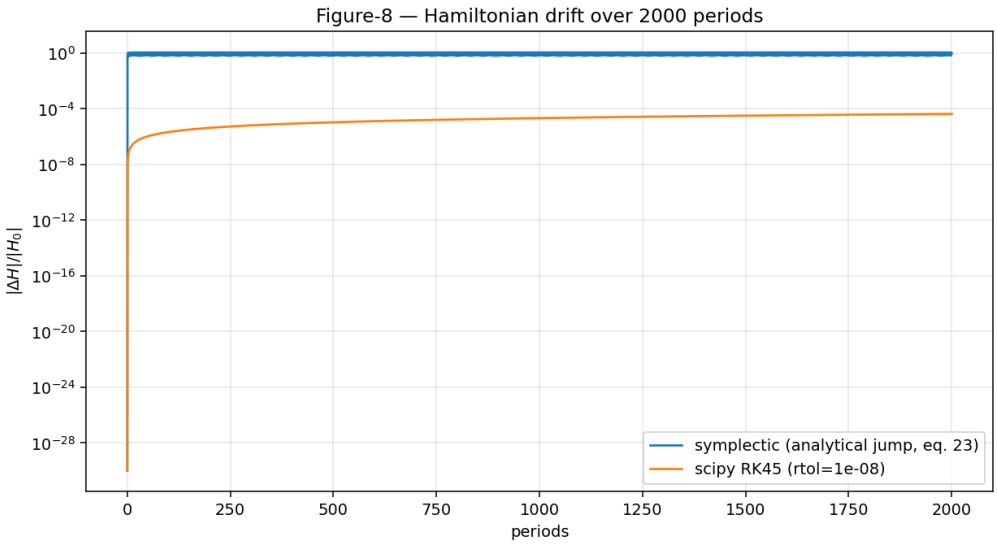


Figure 3: Relative Hamiltonian drift $|\Delta H|/|H_0|$ over the longest benchmarked horizon (2000 orbital periods). The RK45 line rises monotonically with integration time – the well-known symplecticity failure of non-symplectic explicit integrators. The analytical-jump line fluctuates within a bounded envelope set by the exchange of energy between the real-space kinetic terms and the imaginary buffer coordinates. The genuine conserved quantity of the 6D framework is the hyperbolic invariant $(\alpha_s + \beta_s)^2 - M^2 = 4$, which is preserved to machine precision by construction and is not directly visible on this plot.